PROJECT PROPOSAL

Motor Insurance Claim Propensity Prediction

Predicting policyholder claim likelihood at renewal
using Porto Seguro data + World Bank macro indicators

595k	**57**	**3.6%**
Policyholders	Features	Claim rate

The business problem

Blunt pricing today

Insurers use broad actuarial tables — age, car type, region. Fine-grained risk is invisible.

Cost of mispricing

Undercharge high-risk → losses. Overcharge low-risk → customer churn to competitors.

The opportunity

Identify high-risk policyholders at renewal — before the claim happens — to price accurately and intervene proactively.

3–5%

of policyholders

drive

60–80%

of claim costs

Goal: Predict at renewal whether a policyholder will file a claim in the next policy year — enabling actuarial pricing and targeted intervention.

The dataset

ANCHOR — Porto Seguro (Kaggle)

595,212 real policyholder records

- 57 features: individual profile, vehicle registration, car attributes
- Binary label: filed a claim next year (1) or not (0)
- Class imbalance: ~3.6% positive rate → requires SMOTE / threshold tuning
- Real data from Porto Seguro, Brazil's largest motor insurer
- Mix of binary, categorical, and continuous features

join on year →

AUXILIARY — World Bank API

Brazil macroeconomic indicators

- GDP growth rate (annual %)
- Inflation — consumer prices (annual %)
- Unemployment rate (% of labour force)
- Joined to Porto Seguro records at policy year level
- Rationale: macro conditions affect driving frequency and claim behaviour

Users and how they use the system

Actuary / Underwriter

When

At policy renewal (monthly cycle)

Receives

Risk score per policyholder → tiered as Low / Medium / High

Action

Adjusts premium band, flags profile for manual review, or offers telematics programme

Note

Never sees raw model probabilities — only risk tier

Data / Ops Team

When

Ongoing — weekly monitoring

Receives

Model performance dashboard + drift alerts

Action

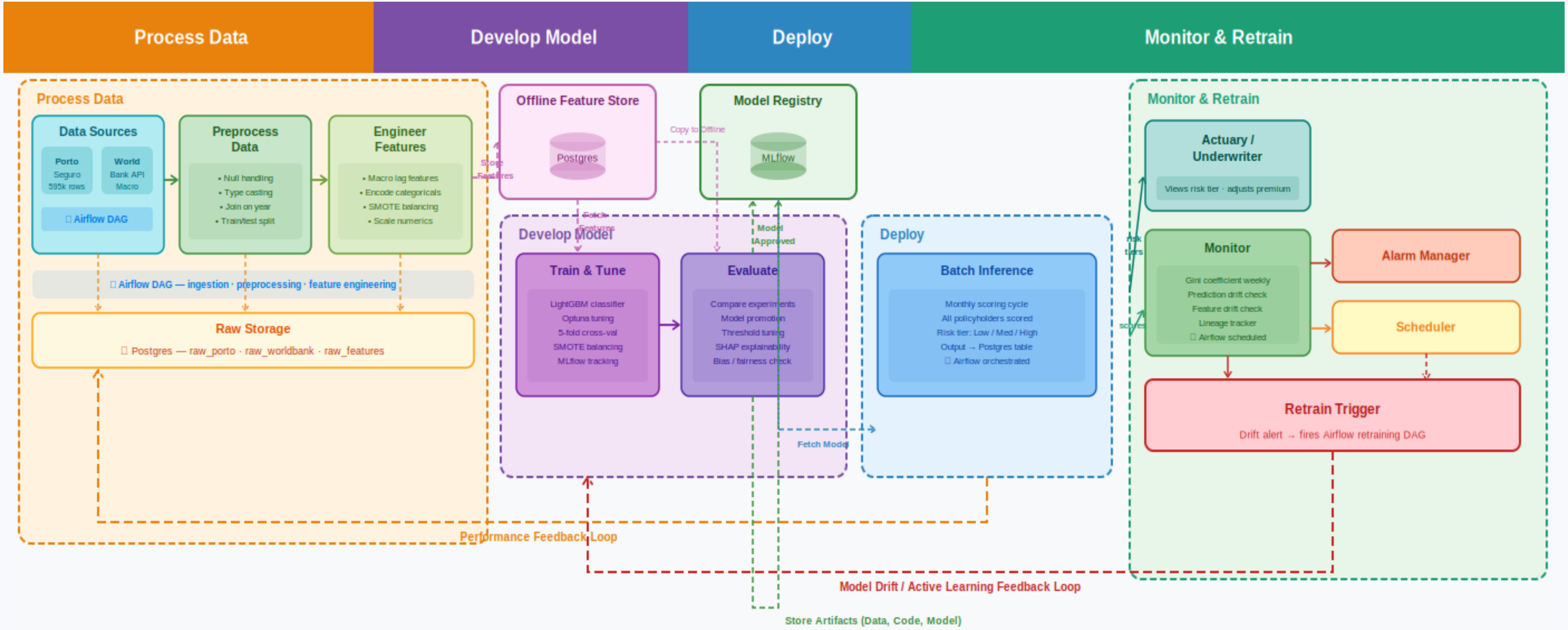
Reviews prediction distribution, triggers retraining pipeline if Gini drops below threshold

Note

Internal tooling only — not customer-facing

System is internal decision-support only — not customer-facing. Output consumed at the renewal workflow stage.

High-level architecture



All components containerised via Docker + docker-compose — runs fully locally · deployable to any cloud with same tooling

Design choices & open questions

Design choices — rationale

Batch over real-time inference

Renewal decisions are monthly — not millisecond. Batch is the right fit.

LightGBM over deep learning

Tabular data, interpretability needed for actuarial review, faster iteration cycle.

SMOTE for class imbalance

3.6% claim rate requires oversampling. Alternative: threshold tuning — open to feedback.

MLflow for experiment tracking

Reproducibility and model versioning are core to production MLOps.

Open questions — seeking feedback

SMOTE vs threshold tuning?

Which approach better suits a class imbalance of 3.6%?

Add SHAP explainability layer?

So underwriters can see why a policyholder was flagged. Worth the added complexity?

Time-based vs drift-based retraining?

Retrain quarterly on schedule, or trigger only when Gini drops below threshold?

Single model vs segmented models?

One global model, or separate models per vehicle category?